

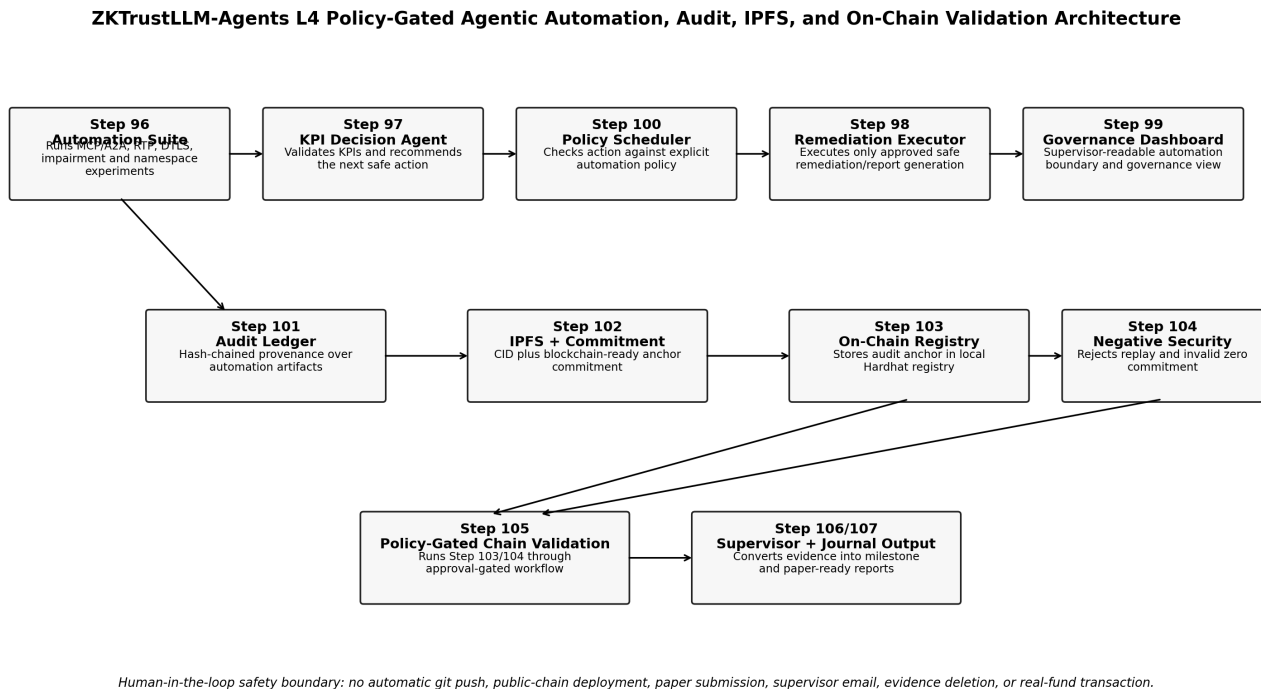
ZKTrustLLM-Agents L4 Journal-Ready Architecture and Result Tables

Steps 96-105: Policy-Gated Agentic Automation, Audit Ledger, IPFS Anchor, and On-Chain Validation

Purpose

This report converts the implemented Level 4 automation and audit chain into journal-ready material: one architecture figure, workflow component table, evidence table, contribution mapping table, and governance-boundary table.

Figure 6.1. Policy-Gated Agentic Automation Architecture



Caption: Policy-gated Level 4 ZKTrustLLM-Agents automation architecture linking automated experiment execution, KPI-based decisioning, human-approved remediation, governance, hash-chained audit provenance, IPFS/blockchain-ready anchoring, local on-chain registry validation, and replay/invalid-anchor negative-security testing.

Table 6.1. Workflow Components

Step	Component	Status	Role
Step 96	Agentic automation suite	PASS	Runs MCP/A2A, media-plane, DTLS, impairment, and namespace experiments.
Step 97	Closed-loop KPI decision agent	GENERATE_SUPERVISOR_REPORT	Reads KPI artifacts and recommends the next safe action.
Step 98	Closed-loop remediation executor	EXECUTED_PASS	Executes approved remediation/report generation with human approval.
Step 99	Automation governance dashboard	JSON_PRESENT	Summarises governance boundaries for supervisor review.
Step 100	Policy-gated autonomous scheduler	EXECUTED_PASS	Connects KPI decisioning and remediation through policy gates.
Step 101	Hash-chained audit ledger	14 entries	Creates tamper-evident artifact provenance.
Step 102	IPFS/blockchain-ready anchor	IPFS_ADD_PASS	Creates IPFS CID and blockchain-ready anchor commitment.
Step 103	On-chain audit-anchor registry	anchorId=0	Stores the audit anchor in a local Hardhat smart-contract registry.
Step 104	Negative-security validation	PASS	Tests replay rejection and invalid zero-commitment rejection.
Step 105	Policy-gated on-chain validation	EXECUTED_PASS	Runs Step 103/104 validation through explicit approval gates.

Table 6.2. Audit and On-Chain Evidence Values

Evidence	Value	Source
Automation status	PASS	Step 96
KPI decision	GENERATE_SUPERVISOR_REPORT	Step 97
Scheduler status	EXECUTED_PASS	Step 100
Audit ledger entries	14	Step 101
Final ledger hash	0db58b3f2c09e530c280dc6cd795a53df54e3337b4a46aa9c70fe6ea166c2567	Step 101
IPFS status	IPFS_ADD_PASS	Step 102
IPFS CID	QmW7VeYs3fP5kk92JstT8tAEkQytgseH4SfhKRpcSwU112	Step 102
Anchor commitment hash	50aa798c5968470f172f5e91cb9d80e25b6ac018d3acde55a424f32ec1c2c8ec	Step 102
On-chain transaction hash	0x3aa06732baa252f303eb24a6b20c118a6a3c665b7e8cbcb6dce0a2b9d3053277	Step 103
On-chain gas used	320095	Step 103
Commitment match check	True	Step 103
Duplicate replay rejected	True	Step 104
Zero commitment rejected	True	Step 104
Policy-gated on-chain validation	EXECUTED_PASS	Step 105

Table 6.3. Journal Contribution Mapping

ID	Claim	Evidence	Journal Meaning
C1	Policy-gated agentic experiment automation	Steps 96-100	Shows repeatable automation with human/supervisor governance boundaries.
C2	Closed-loop KPI decision and remediation	Steps 97-98	Converts KPI evidence into safe next-action recommendations and approved execution.
C3	Tamper-evident automation provenance	Step 101	Hash-chained audit ledger links experiment, decision, remediation, governance, and reports.
C4	IPFS/blockchain-ready audit anchoring	Step 102	Binds the audit ledger to content addressing and a commitment hash.
C5	Smart-contract audit-anchor validation	Steps 103-104	Stores anchor commitments and validates replay/invalid-commitment rejection.
C6	Policy-gated blockchain validation	Step 105	Brings on-chain validation into the controlled automation workflow.

Table 6.4. Governance Boundary

Boundary Class	Actions
Allowed automatically	Read artifacts; validate JSON/CSV; generate reports; create manifests; recommend next action.
Allowed with human approval	Execute remediation; regenerate reports; rerun non-privileged experiments.
Allowed with human and privileged approval	Rerun tc/netem and namespace experiments requiring sudo.
Never automatic	Git push; modify Git history; submit papers; email supervisor; deploy public-chain contracts; delete evidence; spend funds.